

Final Report

The Art of Scientific Computing: Project Title: Opinion Dynamics

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Chapter 1

Introduction

Opinion dynamics sits at the intersection of the social sciences, mathematics and physics - in which the tools of stochastic modelling and statistical physics can be used to study models for the behaviour of people in social systems, such as the dissemination of information and preference in a population [CFL09; Sir+16]. In this paper we consider two models in particular, the threshold voter model and the CODA model, both simulate how opinions propagate in a population. The models differ in that the voter model only allows 'discrete opinions' i.e you either agree or you disagree, all or nothing. The CODA model on the other hand allows for 'continuous opinions' and thus shades of grey. We will implement simulations of these models and conduct some computational experiments in regards to their behaviour. We will consider techniques from statistical mechanics of finding scale invariants and phase transitions to analyse behaviour as we vary parameters.

Chapter 2

The Voter Model

Our first model of opinion dynamics is the threshold voter model [VA18].

2.1 Definitions

Definition 2.1.1 (The Threshold Voter Model)**.** Let $G = (V, E)$ be an undirected graph with vertex set V and edge set E. If $x \in V$ we will write $V(x) \subseteq V$ for the set of vertices with an edge connecting them to x. Fix $T \in \mathbb{Z}_{\geq 0}$. We define a continuous time Markov process with configuration space $C = \prod_{v \in V} \{0, 1\}$ and for a fixed configuration $\eta \in C$ rates given by

$$c(x, \eta) = \begin{cases} 1, & \text{if} \quad |\{y \in V(x) \mid \eta(x) \neq \eta(y)\}| \geq T \\ 0, & \text{otherwise} \end{cases}, \tag{2.1.1}$$

for all $x \in V$. Instead of dealing with this continuous time Markov process directly, it is common to take the discrete time Markov process underlying it. In this paper we will call this underlying discrete time Markov process the threshold voter model. Equivalently this process can be thought of as the procedure of randomly sampling the vertices with more than or equal to T neighbours disagreeing with them and then flipping the value of the sampled vertex.

Definition 2.1.2 (Integer Lattice Graph)**.** We consider the graph with vertices given by the integer lattice $\mathbb{Z}^2$. To define our edges we consider the ℓ^1 metric for $(x, y), (a, b) \in \mathbb{Z}^2$ given by

$$d((x, y), (a, b)) = |x - a| + |y - b| \tag{2.1.2}$$

and say that there is an edge between (x, y) and (a, b) if $d((x, y), (a, b)) \leq 1$, i.e our graph is the standard square integer lattice with an additional reflex edges at each vertex.

2.2 Limiting Behaviour

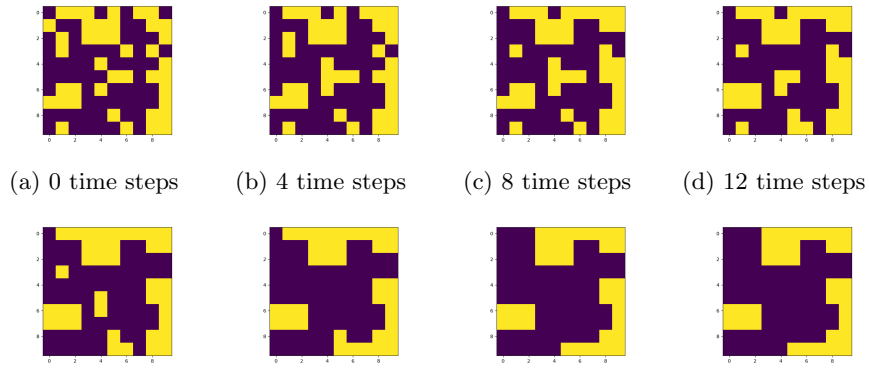
Theorem 2.2.1. Using the integer lattice as our graph if $T > 2$ then each site will change values finitely often, we say the process fixates[HL75].

We will verify this theorem and explore what happens for $T \leq 2$ via an implementation of the voter model in python. Our implementation will be on a lattice defined similarly to Definition 2.1.2 however due to hardware constraints we are unable to simulate an infinite lattice

and instead consider the lattice on the torus, i.e we have a finite $N \times M$ integer lattice and connect the boundaries, so the vertex $(x, 0)$ will be connected to $(x, M-1)$ for all $x \in \{0, \dots N-1\}$ and the vertex $(0, y)$ will be connected to $(N-1, y)$ for all $y \in \{0, \dots, M-1\}$. To choose our initial conditions we uniformly sample a configuration from the configuration space, this will provide us with a noisy initial condition, good for seeing what the generic dynamics are.

We experimentally find that for $T \geq 2$ the process will always end up in a stationary state. We briefly discuss the difference between the dynamics in the $T = 2, 3$ cases.

For $T = 3$ we observe that the dynamics are 'sticky'. That is, since the threshold variable is so high there are many possible stable configurations in the configuration space. Where by stable configuration we mean a configurations which does not evolve with time. This abundance of stable configurations means that the initial configuration does not need to change drastically to reach a stable configuration. We see this qualitatively in Figure 2.1 where some of the features of the initial configuration remain in the final configuration.



(a) 0 time steps (b) 4 time steps (c) 8 time steps (d) 12 time steps

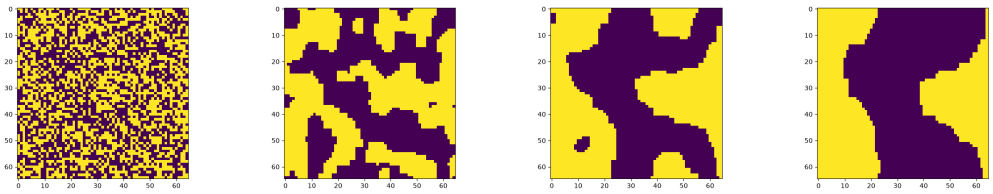
(e) 16 time steps (f) 20 time steps (g) 24 time steps (h) 28 time steps

Figure 2.1: 10×10 grid with $T = 3$ evolving in time steps of size 4.

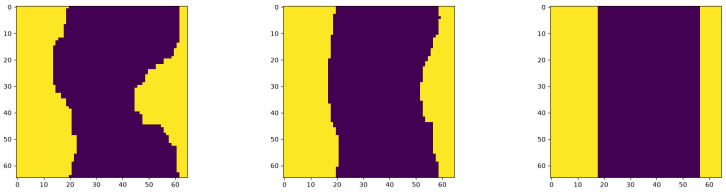
For $T = 2$ the process has far less stable configuration and so the process will evolve to look quite different from its initial configuration. Experimentally the only stable configurations we observed were configurations with solid bands of one opinion crossing the entire lattice, we see an example of such a configuration in Figure 2.2. We also still have the trivial stable configurations of all one opinion.

We will now quantify this difference in behaviour. We consider the statistic given by the average number of changed opinions from the uniformly sampled initial configuration to the final stationary configuration divided by the size of the lattice. We plot this statistic in Figure 2.3 and observe that it is a scale invariant, that is it remains invariant under scaling the lattice. We see that as T increase the change from initial to final configuration decreases rapidly until at $T \geq 5$ the process is static.

We omit the $T \leq 1$ case in the above discussion as for this choice the process no longer fixates. This can be seen in Figure 2.4 where even after substantial time steps the configuration is still qualitatively just noise.



(a) 0 time steps (b) 7,750 time steps (c) 15,500 time steps (d) 23,250 time steps



(e) 31,000 time steps (f) 38,750 time steps (g) 46,500 time steps

Figure 2.2: 65×65 grid with $T = 2$ evolving in steps of 7750.

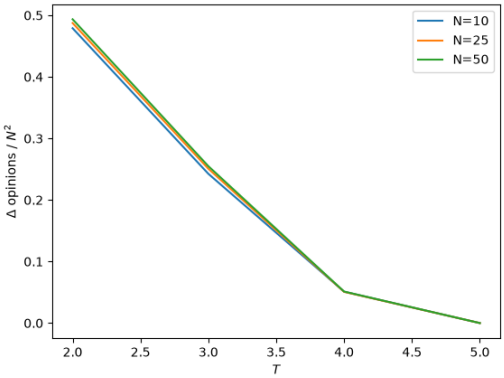
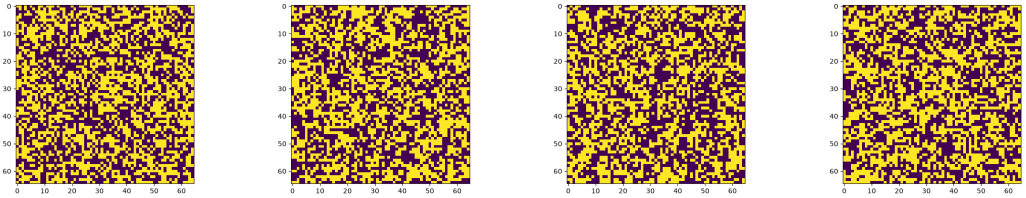


Figure 2.3: We consider an $N \times N$ lattice and average over 25 randomly sampled initial configurations.



(a) 0 time steps (b) 7,750 time steps (c) 15,500 time steps (d) 23,250 time steps

Figure 2.4: 65×65 grid with $T = 1$ evolving in steps of 7750.

Chapter 3

CODA Model

3.1 definition

The CODA model [Mar08] is a process on a graph, $G = (V, E)$, where we associate a probability to each vertex. We follow [Mar10] for the definition of a generalisation of the CODA model. We imagine we have two mutually exclusive outcomes A, B and at each vertex v we associate a probability $p_v \in [0, 1]$ where p_v is thought of as the probability that v believes in outcomes A and thus $1 - p_v$ is the probability that v believes in outcome B. We associate an action to each vertex, notated $\sigma_v \in \pm 1$ with $\sigma_v = 1$ if $p_v \geq \frac{1}{2}$ and $\sigma_v = -1$ if $p_v < \frac{1}{2}$. We then define the following constants

$$\begin{aligned} a &= P_v(\sigma_u = 1 \mid A, \sigma_v = 1), & P_v(\sigma_u = 1 \mid A, \sigma_v = -1) &= c, \\ b &= P_v(\sigma_u = -1 \mid B, \sigma_v = -1), & P_v(\sigma_u = -1 \mid B, \sigma_v = 1) &= d. \end{aligned} \tag{3.1.1}$$

for all vertices $u, v \in V$, we assume that these are in fact constants not depending on u, v. Here P_v refers to the probability distribution at site v i.e $P_v(A) = p_v$. This assumption is saying that we are not providing any kind of 'spatial bias' in our model. We then update our model by randomly sampling a vertex $v \in V$ and a neighbour $u \in V(v)$ then using Bayes' rule in the following way to get an updated probability at vertex v denoted p'_v. We have different cases depending on the opinion of the neighbour and the opinion of the vertex itself.

If $\sigma_u = 1$ and $\sigma_v = 1$ then

$$\begin{aligned} p'_v = P_v(A \mid \sigma_u = 1, \sigma_v = 1) &= \frac{P_v(\sigma_u = 1 \mid A, \sigma_v = 1) P_v(A \mid \sigma_v = 1)}{P_v(\sigma_u = 1 \mid \sigma_v = 1)} \\ &= \frac{a p_v}{P_v(\sigma_u = 1 \mid \sigma_v = 1)}, \end{aligned} \tag{3.1.2}$$

and

$$\begin{aligned} 1 - p'_v = P_v(B \mid \sigma_u = 1, \sigma_v = 1) &= \frac{P_v(\sigma_u = 1 \mid B, \sigma_v = 1) P_v(B \mid \sigma_v = 1)}{P_v(\sigma_u = 1 \mid \sigma_v = 1)} \\ &= \frac{(1 - d)(1 - p_v)}{P_v(\sigma_u = 1 \mid \sigma_v = 1)}. \end{aligned} \tag{3.1.3}$$

If $\sigma_u = -1$ and $\sigma_v = 1$ then

$$\begin{aligned} p'_v = P_v(A \mid \sigma_u = -1, \sigma_v = 1) &= \frac{P_v(\sigma_u = -1 \mid A, \sigma_v = 1) P_v(A \mid \sigma_v = 1)}{P_v(\sigma_u = -1 \mid \sigma_v = 1)} \\ &= \frac{(1-a)p_v}{P_v(\sigma_u = 1 \mid \sigma_v = 1)}, \end{aligned} \tag{3.1.4}$$

and

$$\begin{aligned} 1 - p'_v = P_v(B \mid \sigma_u = -1, \sigma_v = 1) &= \frac{P_v(\sigma_u = -1 \mid B, \sigma_v = 1) P_v(B \mid \sigma_v = 1)}{P_v(\sigma_u = -1 \mid \sigma_v = 1)} \\ &= \frac{d(1-p_v)}{P_v(\sigma_u = 1 \mid \sigma_v = 1)}. \end{aligned} \tag{3.1.5}$$

If $\sigma_u = 1$ and $\sigma_v = -1$ then

$$\begin{aligned} p'_v = P_v(A \mid \sigma_u = 1, \sigma_v = -1) &= \frac{P_v(\sigma_u = 1 \mid A, \sigma_v = -1) P_v(A \mid \sigma_v = -1)}{P_v(\sigma_u = 1 \mid \sigma_v = -1)} \\ &= \frac{cp_v}{P_v(\sigma_u = 1 \mid \sigma_v = -1)}, \end{aligned} \tag{3.1.6}$$

and

$$\begin{aligned} 1 - p'_v = P_v(B \mid \sigma_u = 1, \sigma_v = -1) &= \frac{P_v(\sigma_u = 1 \mid B, \sigma_v = -1) P_v(B \mid \sigma_v = -1)}{P_v(\sigma_u = 1 \mid \sigma_v = -1)} \\ &= \frac{(1-b)(1-p_v)}{P_v(\sigma_u = 1 \mid \sigma_v = 1)}. \end{aligned} \tag{3.1.7}$$

If $\sigma_u = -1$ and $\sigma_v = -1$ then

$$\begin{aligned} p'_v = P_v(A \mid \sigma_u = -1, \sigma_v = -1) &= \frac{P_v(\sigma_u = -1 \mid A, \sigma_v = -1) P_v(A \mid \sigma_v = -1)}{P_v(\sigma_u = -1 \mid \sigma_v = -1)} \\ &= \frac{(1-c)p_v}{P_v(\sigma_u = -1 \mid \sigma_v = -1)}, \end{aligned} \tag{3.1.8}$$

and

$$\begin{aligned} 1 - p'_v = P_v(B \mid \sigma_u = -1, \sigma_v = -1) &= \frac{P_v(\sigma_u = -1 \mid B, \sigma_v = -1) P_v(B \mid \sigma_v = -1)}{P_v(\sigma_u = -1 \mid \sigma_v = -1)} \\ &= \frac{b(1-p_v)}{P_v(\sigma_u = -1 \mid \sigma_v = -1)}. \end{aligned} \tag{3.1.9}$$

To avoid having to deal with the terms on the denominators of the above expressions it is convention to consider the odds at a given site instead of a probability. For example when $\sigma_u = 1$ and $\sigma_v = 1$ then

$$\frac{p'_v}{1 - p'_v} = \frac{a}{1-d} \frac{p_v}{1-p_v}. \tag{3.1.10}$$

Additionally to make our computations easier one considers the *log-odds* at a given site by taking the log of the odds. We write η_v for the log-odds at a site v and have the update rule where η'_v is the updated log odds

$$\eta'_v = \begin{cases} \eta_v + \log \frac{a}{1-d}, & \text{if} \quad \sigma_u = 1, \sigma_v = 1 \\ \eta_v - \log \frac{d}{1-a}, & \text{if} \quad \sigma_u = -1, \sigma_v = 1 \\ \eta_v + \log \frac{c}{1-b}, & \text{if} \quad \sigma_u = 1, \sigma_v = -1 \\ \eta_v - \log \frac{b}{1-c}, & \text{if} \quad \sigma_u = -1, \sigma_v = -1 \end{cases}. \tag{3.1.11}$$

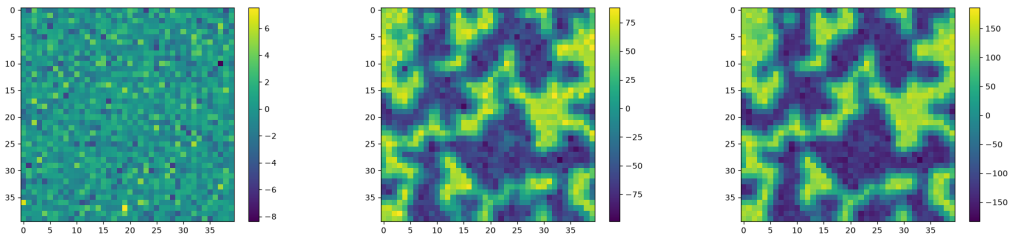
We summarise the process as follows. We have at each vertex $v \in V$ an associated log-odds $\eta_v \in (-\infty, \infty)$. To evolve this system we first uniformly sample our vertex set $v \in V$ and its neighbourhood $u \in V(v)$, we then apply the update rule Equation (3.1.11) to update the log-odds at v. Note that in this log-odds setting, $\sigma_v = 1$ if $\eta_v \geq 0$ and $\sigma_v = -1$ if $\eta_v < 0$.

We would like to think of this model in the context of opinion dynamics with this update rule representing an interaction between site v and its neighbour site u with site u imparting some of its opinion on site v. In this context it is physical to set $a + d \geq 1$ and $c + d \geq 1$, as this ensures that the update rule will cause the preference of site v to move in the direction of the preference of site u after an interaction.

3.2 Dynamics

As we did in Section 2.2 we will simulate this process on a graph corresponding to an $N \times N$ integer torus, however unlike in Definition 2.1.2 we will define our edges by $d(x, y) = 1$ i.e excluding the reflexive edge going from each vertex to itself. To sample an initial conditions we uniformly sample a probability $p_v \in [0, 1]$ for each vertex $v \in V$ in the lattice, and then transform these initial probabilities into the log-odds picture.

We see the dynamics in an 'unbiased' regime in Figure 3.1 where we take $a = b = c = d = \frac{3}{4}$. We qualitatively observe that regions of positive log-odds and regions of negative log-odds emerge with borders of 'undecided' vertices with opinions close to zero in between the regions. We can



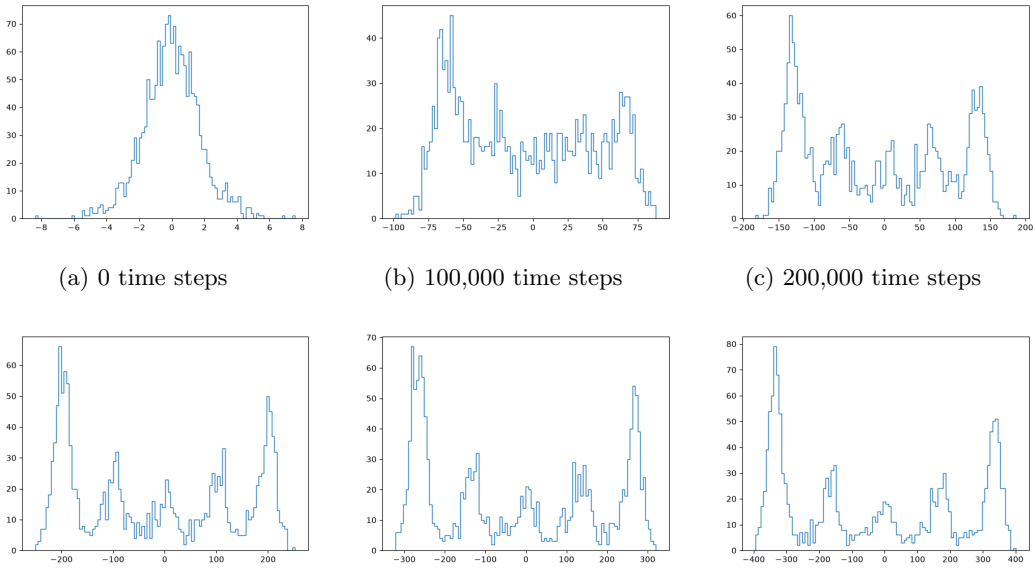
(a) 0 time steps (b) 100,000 time steps (c) 200,000 time steps

(d) 300,000 time steps (e) 400,000 time steps (f) 500,000 time steps

Figure 3.1: Taking $a = b = c = d = \frac{3}{4}$ evolving in time steps of 100,000 from left to right. Each subfigure is a heat-map of the log-odds on the lattice.

further examine this behaviour in Figure 3.2 in which we have histograms for the log-odds of the configurations in Figure 3.1. We see that as system evolves the log-odds spread out from the initial distribution around 0 and then collect into a set of peaks symmetric around 0. We have

two peaks of extreme opinions, two peaks of more moderate opinions and a peak of near zero opinions. Before we continue with further discussion of this behaviour we consider four other



(a) 0 time steps (b) 100,000 time steps (c) 200,000 time steps

(d) 300,000 time steps (e) 400,000 time steps (f) 500,000 time steps

Figure 3.2: Same configurations as Figure 3.1 however here we plot a histogram of log-odds.

distinct regimes with which we can compare.

The four regimes we will consider are as follows, under the example of a vertex $v \in V$ and neighbour $u \in V(v)$.

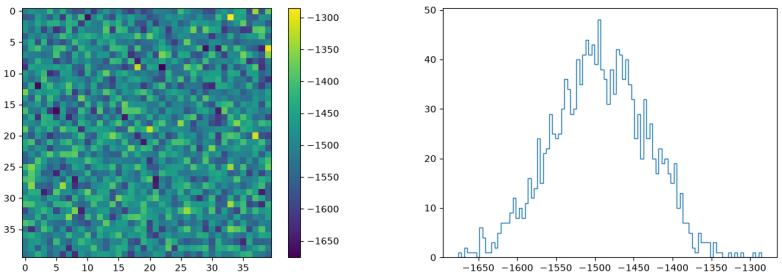
(a) Setting $a = c > b = d$ we have that the *negative direction is favoured*. In this regime the change of the opinion of v is independent of σ_v and the change when $\sigma_u = -1$ is larger in magnitude compared to when $\sigma_u = 1$.

(b) Setting $a = b > c = d$ we have that *disagreement is favoured*. In this regime the magnitude of the change of the opinion of v is only dependent on if $\sigma_u \neq \sigma_v$ or $\sigma_u = \sigma_v$, i.e the magnitude depends only on agreement vs disagreement with their neighbour. Further the magnitude of the change when $\sigma_u \neq \sigma_v$ is larger then when $\sigma_u = \sigma_v$.

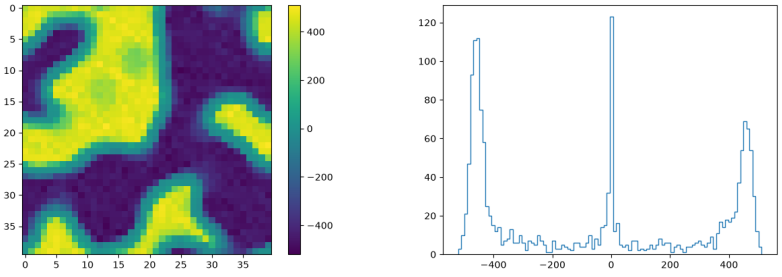
(c) Setting $a = c < b = d$ we have that the *positive direction is favoured*. This regime has the same independence of regime (a) however the change when $\sigma_u = 1$ is larger in magnitude compared to when $\sigma_u = -1$.

(d) Setting $a = b < c = d$ we have that *agreement is favoured*. This regime has the same independence of regime (b) however the magnitude of the change when $\sigma_u = \sigma_v$ is larger then when $\sigma_u \neq \sigma_v$.

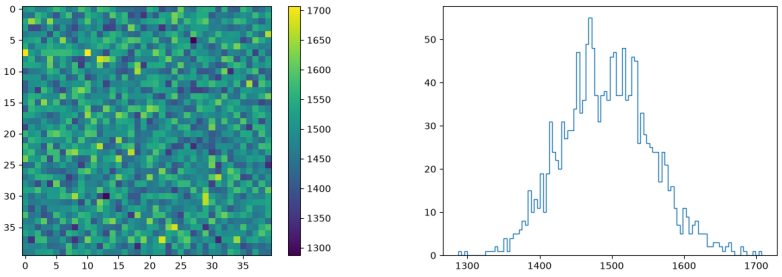
We see examples of how each of these regimes behave in Figure 3.3. We acknowledge that these regimes do not cover all choices of our variables however they provide us with a good look at the different kinds of behaviour this process can exhibit.



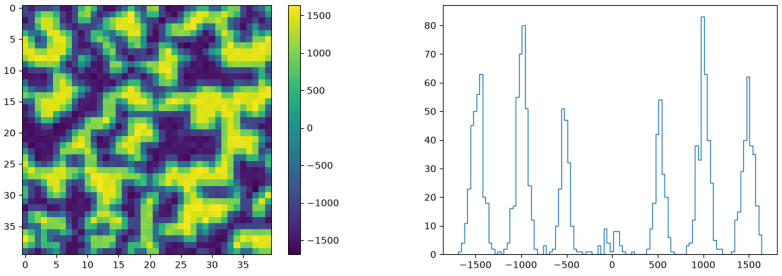
(a) Favouring negative, $a = c = 19/20$, and $b = d = 11/20$.



(b) Favouring disagreement, $a = b = 19/20$, and $c = d = 11/20$.



(c) Favouring positive, $a = c = 11/20$, and $b = d = 19/20$.



(d) Favouring agreement, $a = b = 11/20$, and $c = d = 19/20$.

Figure 3.3: Four extreme regimes after 1,000,000 time steps. On the left a heat map of the configuration and on the right a histogram of the log-odds.

In regimes (a) and (c) the heat maps are qualitatively still just noise however they have become very negative or very positive respectively. We can explain this behaviour as follows. We consider the transformed process for $v \in V$ after t time steps,

$$\zeta_v(t) = \eta_v(t) - \frac{t}{2}\left(\log\frac{a}{1-d} - \log\frac{d}{1-a}\right), \tag{3.2.1}$$

where $\eta_v(t)$ is the standard CODA process after t time steps. So the update rule for ζ is for $v \in V$ and $u \in V(v)$,

$$\begin{aligned}\zeta_v(t+1) - \zeta(t) &= \eta(t+1) - \eta(t) - \frac{1}{2}\left(\log\frac{a}{1-d} - \log\frac{d}{1-a}\right) \\ &= \begin{cases} \frac{1}{2}\left(\log\frac{a}{1-d} + \log\frac{d}{1-a}\right), & \text{if} \quad \sigma_u = 1 \\ -\frac{1}{2}\left(\log\frac{a}{1-d} + \log\frac{d}{1-a}\right), & \text{if} \quad \sigma_u = -1 \end{cases}\end{aligned} \tag{3.2.2}$$

which can be rewritten as

$$\zeta_v' = \begin{cases} \zeta_v + \frac{1}{2}\left(\log\frac{a}{1-d} + \log\frac{d}{1-a}\right), & \text{if} \quad \sigma_u = 1 \\ \zeta_v - \frac{1}{2}\left(\log\frac{a}{1-d} + \log\frac{d}{1-a}\right), & \text{if} \quad \sigma_u = -1 \end{cases}. \tag{3.2.3}$$

At first glance one would think that this is equivalent under rescaling of variables to the unbiased case we introduced earlier in which $a = b = c = d$. However recall that the value of σ_u is dependent on the sign of η_u since we have a drift in either the positive or negative direction after enough time we conject that all vertices will become negative and thus the process will always evolve by moving in the biased direction. This is why we see the histogram we do, initial the process is able to spread somewhat as it has similar dynamics to the unbiased case however after enough time the drift causes all vertices to have the same sign at which point the shape of the histogram becomes qualitatively static and the process continuous to drift away from 0.

In regimes (b) and (d) we have the same symmetry around 0 that we observed in the unbiased case. For regime (b) we observe qualitatively that the regions of agreement seem to be larger and the boundaries between these regions thicker. The latter of which is manifest in our histogram in which there is a much higher peak at 0 compared to the unbiased case. We also see that there is only two peaks of non-zero log-odds compared to the four peaks we had in the unbiased case. This is quite different to the behaviour in regime (d) in which we have much slimmer regions of agreement with very thin boundaries between them. In the histogram the peak at 0 is almost gone and there are a set of six non-zero peaks.

3.3 Phase Transition

In this section we will consider a statistic with which we can further analyse this models.

3.3.1 ℓ^1 Distance

Recall in Definition 2.1.2 we defined a metric, d, on the square integer lattice $\mathbb{Z}^2$. We can create an analogous metric on the square integer torus $\{0, \ldots, N-1\}^2$ with $N \in \mathbb{N}$ as follows. For $(x, y), (a, b) \in \{0, \ldots, N-1\}^2$ we then define

$$d(x, y) = \min\{|x-a|,\, N - |x-a|\} + \min\{|y-b|,\, N - |y-b|\}. \tag{3.3.1}$$

In words this distance will give us the smallest number of edges needed to traverse from a vertex $v \in \{0, \dots, N-1\}^2$ to another vertex $u \in \{0, \dots, N-1\}^2$ on the integer torus graph. For a configuration η we define the statistic

$$B_\eta = \frac{1}{N^2} \sum_{v \in \{0,\dots,N-1\}^2} \min\left(\{d(v,u) \mid \sigma_v \neq \sigma_u\} \cup \{2N\}\right). \tag{3.3.2}$$

B_η is the distance from a vertex v to the closest vertex u which disagrees with v, averaged over all v on the lattice. We add $2N$ to the set we are taking the minimum over in case all vertices on the lattice agree, as this will force $B_\eta = 2N$, noting that this is strictly larger then it could be if there was any disagreement in η. To examine how this statistic is dependent on the parameters of the process we consider the following continuous curve in the parameter space for $t \in (0, \frac{1}{2})$

$$(a, b, c, d) = \gamma(t) = \left(t + \frac{1}{2}, \max\{1-t, t+\frac{1}{2}\}, \min\{1-t, t+\frac{1}{2}\}, 1-t\right). \tag{3.3.3}$$

One can see that for $t = \frac{1}{4}$ this curve lies in the unbiased regime, for $t < \frac{1}{4}$ it is in the favouring negative regime, and for $t > \frac{1}{4}$ it is in the favouring disagreement regime. We then consider B_η after 100,000 time steps averaged over samples of initial conditions in Figure 3.4

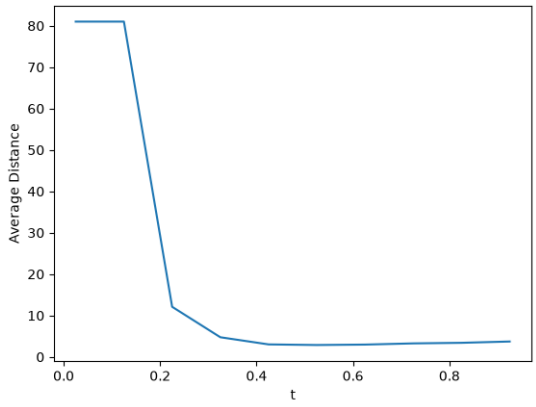


Figure 3.4: Averaged over 10 randomly sampled initial conditions with grid size 40×40.

We see that for $t > \frac{1}{4}$ we have that the average B_η is quite low, less then 5. This agrees with our intuition, as for these values of t we have that the process will be in the favouring disagreement regime and thus we expect there to be a mix of vertices with positive and negative log-odds, giving a relatively low B_η. For $t < \frac{1}{4}$ since we are in the favouring negative regime we expect B_η to be at $2N = 80$, as after enough time steps all vertices should agree. To the contrary we observe that this jump does not in fact kick in until the $t = 0.2$ mark. We explain this as follows. As we approach the unbiased regime from within the favouring negative regime we expect the time it takes for a given state to be in agreement to increase and approach infinity. Thus the jump to $2N$ occurs a bit earlier then we expect, as our simulation only runs for a finite time. We can observe some evidence for this as in Figure 3.5 we do the same plot but taking 1,000,000 time steps. Due to computation limitations we do not average over multiple initial configurations for this plot. We see that the jump has moved to $t = 0.3$, closer to the expected $t = 0.5$ value.

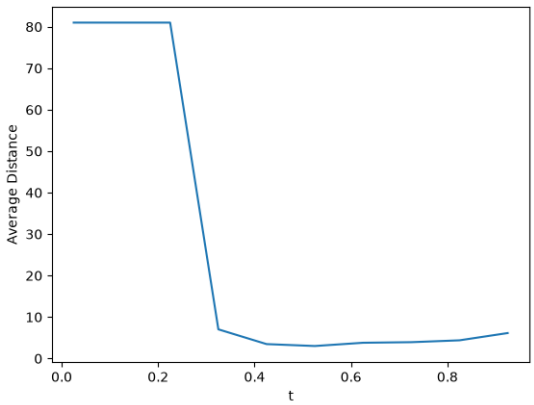


Figure 3.5: For a fixed initial condition with grid size 40×40.

3.4 Variant Model

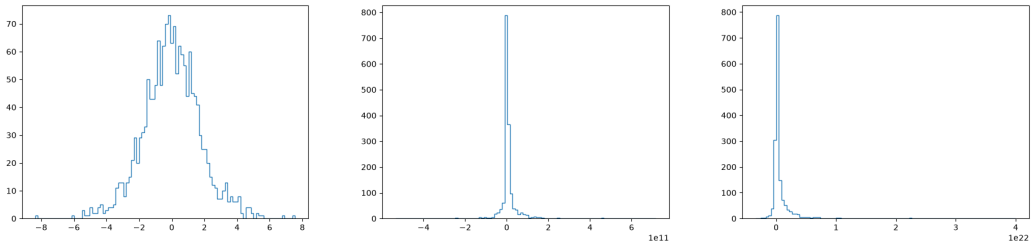
Here we define a modification to the coda model introduced in [Mar10] and replicate the observations made by Martins. In his paper he considers a process where a log-odds configuration η is updated according to the rule

$$\eta'_v = \eta_v + \eta_u \tag{3.4.1}$$

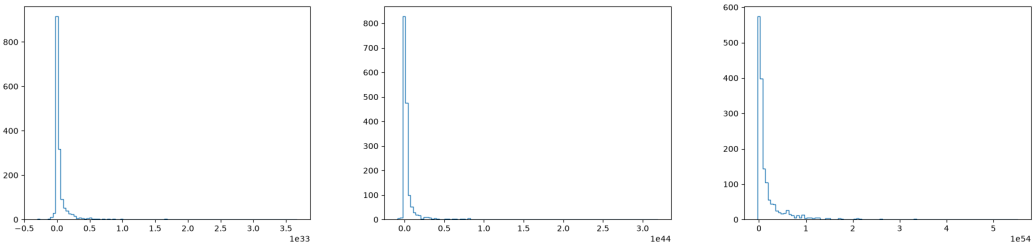
for uniformly sampled $v \in V$ and neighbour $u \in V(v)$. This definition aims to reflect the idea that someone who is less sure about their beliefs, corresponding to being closer to zero, are less effective in convincing others of said beliefs.

3.4.1 Behaviour

We agree with the results of [Mar10] concluding that after sufficiently many time steps any initial configuration will reach consensus. We observe this in Figure 3.6 where after 200,000 time steps all vertices have a positive log-odd associated with them. It seems that as time progresses vertices with extreme opinions appear and begin to dominate the lattice, causing the rest of the lattice to get brought into consensus with them.



(a) 0 time steps (b) 40,000 time steps (c) 80,000 time steps



(d) 120,000 time steps (e) 160,000 time steps (f) 200,000 time steps

Figure 3.6: Histogram of log-odds from a randomly sampled initial configuration over time.

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